

Chapter Three

Security Techniques

Part – II

Substitution Cipher

Information Security

(SE7431)

SUBSTITUTION CIPHER

- ✧ A substitution technique is one in which the letters of plaintext are replaced by other letters, numbers or symbols.
- ✧ If the plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with cipher text bit patterns.

✧ **Mono alphabetic or Simple cipher:**

- ☞ Caesar (Additive) cipher
- ☞ Multiplicative Cipher
- ☞ Affine Cipher

✧ **Polyalphabetic cipher**

- ☞ Vigenère cipher
- ☞ Play fair cipher
- ☞ Hill Cipher
- ☞ Vernam Cipher

SUBSTITUTION CIPHER - MONO ALPHABETIC

- ✧ In mono alphabetic substitution, the relationship between a symbol in the plaintext to a symbol in the cipher text is always one-to-one.
- ✧ Plaintext characters are substituted by a different alphabet stream of characters shifted to the right or left by n positions
- ✧ E.g. ABCDEFGHIJKLMNOPQRSTUVWXYZ = DEFGHIJKLMNOPQRSTUVWXYZABC

✧ Additive Cipher:

- ✧ The simplest mono alphabetic cipher and sometimes called Shift or Caesar cipher, but Additive cipher reveals its mathematical nature
- ✧ Caesar cipher corresponds to $n = 3$

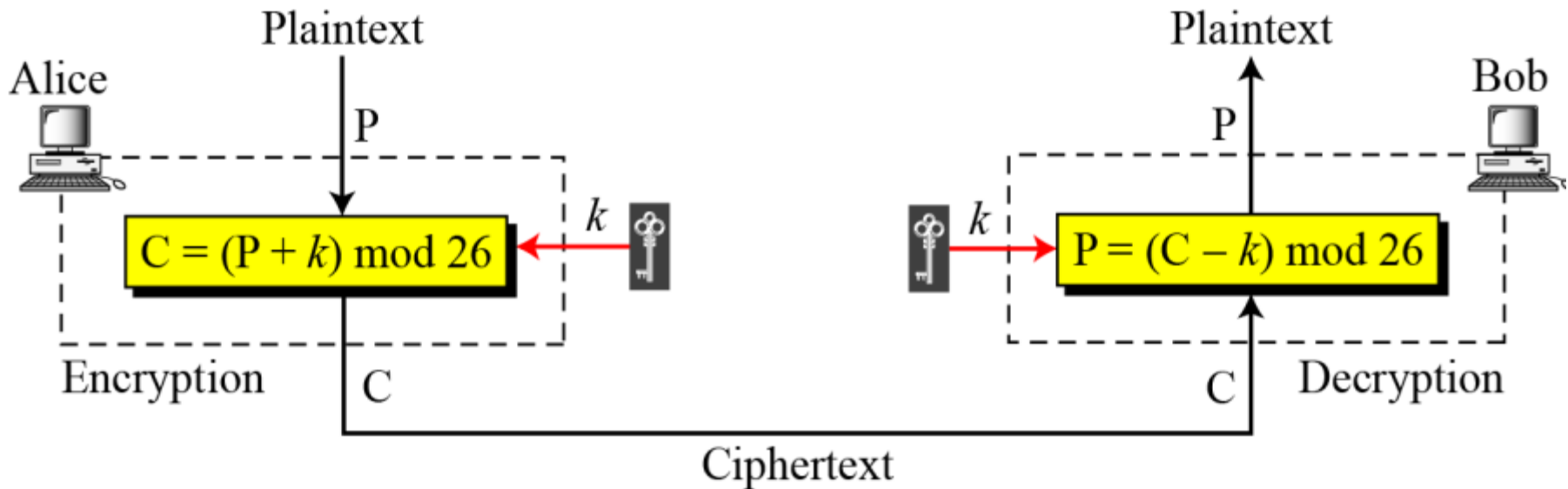
Num.Val	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Plain Text	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Key = 3	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- ✧ Example: **Move the Software Engineers “UP”. (Use Caesar Cipher)**
- ✧ Encryption/Transposition Result: ***PRYH WKH VRIWZDU HQJLQHHUV XS***

SUBSTITUTION CIPHER – MONO ALPHABETIC

🐦 Additive Cipher

- 👉 When the cipher is additive, the plaintext, cipher text, and key are integers in Z_{26}
- 👉 General architecture:



- 👉 Example: Use the additive cipher with key = 15 to encrypt the message “hello”
- 👉 Encryption/Transposition Result: **WTAAD**

SUBSTITUTION CIPHER – MONO ALPHABETIC

✚ Additive Cipher Brute-force Cryptanalysis Attack

- ☞ If it is known that a given cipher text is a Caesar cipher, then a brute-force cryptanalysis is easily performed: simply try all the 25 possible keys.
- ☞ Three important characteristics of this problem enabled us to use a brute-force cryptanalysis:
 - ☞ The encryption and decryption algorithms are known.
 - ☞ There are only 25 keys to try
 - ☞ The language of the plaintext is known and easily recognizable

✚ Brute-force cryptanalysis impractical in an algorithm that employs a large number of keys and if the language of the plain text is unknown

✚ Example: Eve has intercepted the cipher text “PHHW PH DIWHU WKH WRJD SDUWB”.

✚ Encryption/Transposition Result.: **Meet Me After Toga Party**

SUBSTITUTION CIPHER – MONO ALPHABETIC

✧ Multiplicative Cipher

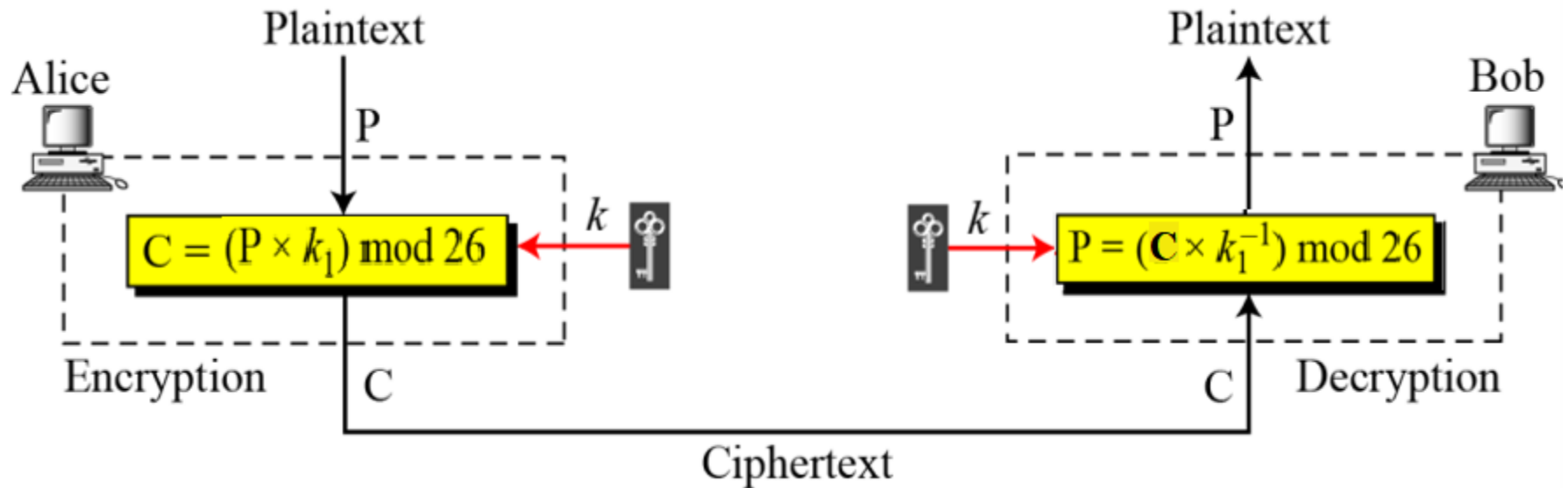
- ☞ Caesar ciphers are encrypted by adding modulo 26 ($C = p + \text{key} \bmod 26$, where C is ciphertext and p is plaintext) and are decrypted by adding the inverse of the key.
- ☞ It seems reasonable to consider what would happen if we encrypted by multiplying modulo 26; i.e., $C = mp \bmod 26$ where m is called the multiplicative key.
- ☞ But, things do not go as well as they did for Caesar ciphers.
- ☞ The plaintext and cipher text are integers in Z_{26} , and the key is also an integer in Z_{26}^*
- ☞ The key domain for any multiplicative cipher: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25
- ☞ Example: Use a multiplicative cipher to encrypt the message “hello” with a key of 7
- ☞ Encryption/Transposition Result: **XCZZU**

✧ Why even numbers in Z_{26}^* are not in the key set of multiplicative cipher?

✧ The only multipliers that are possible are those that result in one-to-one mappings.

SUBSTITUTION CIPHER – MONO ALPHABETIC

✉ Multiplicative Cipher



SUBSTITUTION CIPHER – MONO ALPHABETIC

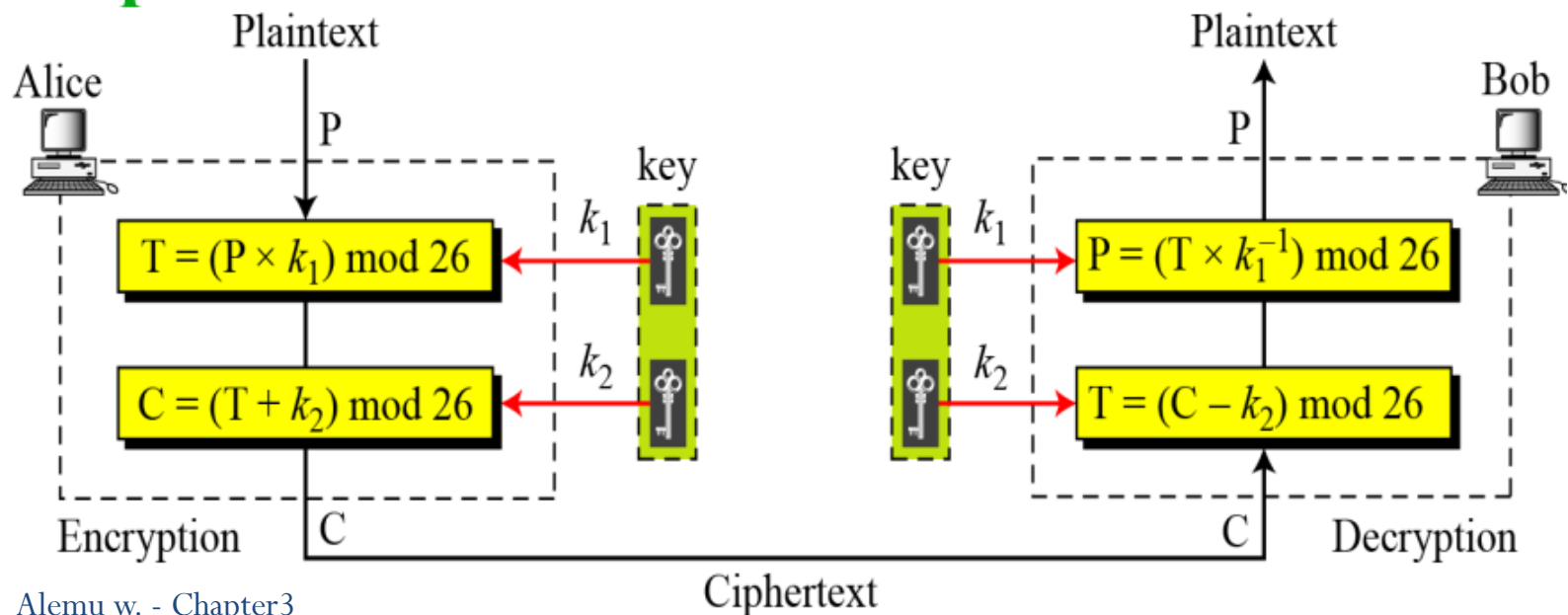
Affine Cipher

- ☞ If encrypting once is good, twice must be better. Correct? Well, not always.
- ☞ Consider encrypting a message twice with Caesar ciphers. Let's say the message was first encrypted using a Caesar cipher with key 7 and then with 11, the result of the re-encryption is equivalent to having encrypted the message once with key 18.
- ☞ Consider encrypting a message twice with multiplicative ciphers. If a message is first encrypted with a key 3 and then with key 7, the re-encryption is equivalent to having encrypted the message once with key 21
- ☞ In neither of these cases is the security enhanced by re-encryption.
- ☞ Well, how about encrypting first with a Caesar cipher and then re-encrypting with a multiplicative cipher? This composition does increase security. Such ciphers are called affine ciphers.

SUBSTITUTION CIPHER – MONO ALPHABETIC

🐦 Affine Cipher:

- We could first encrypt using a multiplicative cipher with multiplicative key m and then re-encrypt with a Caesar cipher with additive key b . This results in $C = mp + b$ where p is plaintext and C is ciphertext.
- Alternatively, we could first encrypt using a Caesar cipher with additive key b and then re-encrypt with a multiplicative cipher with multiplicative key m . This results in $C = m(p + b)$.



SUBSTITUTION CIPHER – MONO ALPHABETIC

✚ Affine Cipher

- ✚ We will agree to always do the multiplicative cipher first
- ✚ Example: Use affine cipher to encrypt the message “hello” with the key (7, 2)
- ✚ Encryption Result: **ZEBBW**

✚ **Example: Use affine cipher to decrypt the message “ZEBBW” with the key pair (7, 2) in modulus 26.**

✚ *Conclusion – Mono Alphabetic Cipher*

- ✚ *Each character was substituted by a single character*
- ✚ *They are not secure enough – operates on the key range of Z_{26} and increasing the key space is required to achieve better security*
- ✚ *Cryptanalysts are helped by the fact that they have to see what character would correspond in plaintext for a given cipher text character*

SUBSTITUTION CIPHER - POLYALPHABETIC

- ✚ Polyalphabetic cipher's goal is to make the process of finding a pattern difficult
- ✚ Each plaintext character may be replaced by more than one character and each occurrence of a character may have a different substitute.
- ✚ Alphabets 'A' through 'Z' are represented by 00, 01, 02, ... , 25
- ✚ The relationship between a character in the plaintext to a character in the cipher text is one-to-many.
- ✚ The most common examples include:
 - ✚ Vigenère cipher
 - ✚ Play fair cipher
 - ✚ Vernam Cipher and
 - ✚ Hill Cipher

SUBSTITUTION CIPHER - POLYALPHABETIC

🕊️ Vigenère cipher

- ☞ Is the most commonly used polyalphabetic cipher method
- ☞ Starts with a 26 x 26 matrix of alphabets in sequence
- ☞ First row starts with 'A', second row starts with 'B', etc.
- ☞ It requires a keyword that the sender and receiver know ahead of time
- ☞ Each character of the message is combined with the characters of the keyword to find the cipher text character
- ☞ Example: Encrypt a message "SEE ME IN MALL." with a keyword INFOSEC
- ☞ *Refer the Vigenère table for your operation [[click here](#)][Or look it at slide 14]*
- ☞ Encryption Result: **ARJAWMPUNQZ**

S	E	E	M	E	I	N	M	A	L	L
I	N	F	O	S	E	C	I	N	F	O

SUBSTITUTION CIPHER - POLYALPHABETIC

Vigenère cipher

- ☞ Easiest way to handle Vigenère cipher is to use arithmetic modulo 26
- ☞ The keyword and message characters are converted to corresponding numbers in the alphabet and added to modulo 26
- ☞ Best feature is that same plaintext character is substituted by different cipher text characters
- ☞ Uses the fact that the keyword character helps to get different cipher text characters from the table
- ☞ Vigenère cipher can be seen as combinations of m additive ciphers
- ☞ To decrypt, use the table, choose the row corresponding to the keyword character and look for the cipher text character in that row
- ☞ Plaintext character is then at the top of that column

SUBSTITUTION CIPHER - POLYALPHABETIC

🦋 Vigenère cipher

☞ Example: Given P = *SHE IS LISTENING* and K = *PASCAL*

☞ Process:

S H E I S L I S T E N I N G

P A S C A L P A S C A L A P

☞ Result: *HHWKSWSLGNPCG*

*Can you do it with
Additive Method?*

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

SUBSTITUTION CIPHER - POLYALPHABETIC

✉ *Multiple Letter (Play fair) Cipher:*

- ✉ Play fair cipher is a multiple letter cipher
- ✉ Each plaintext letter is replaced by a diagram in this cipher
- ✉ Number of possible diagrams is $26 \times 26 = 676$
- ✉ User chooses a keyword and puts it in the cells of a 5 x 5 matrix.
- ✉ I and J stay in one cell, Duplicate letters appear only once.
- ✉ Alphabets that are not in the keyword are arranged in the remaining cells from left to right in successive rows in ascending order
- ✉ To decrypt, the receiver reconstructs the 5 x 5 matrix using the keyword and then uses the same rules as for encryption
- ✉ Example: Example:- Plaintext: “CRYPTO ISTOO EASY” and Keyword is “INFOSEC”

SUBSTITUTION CIPHER

🦋 *Multiple Letter (Play fair) Cipher:*

☞ Grouped text: CR YP TO IS TO XO EA SY

I/J	N	F	O	S
E	C	A	B	D
G	H	K	L	M
P	Q	R	T	U
V	W	X	Y	Z

☞ Result: *AQVTYB NIYBYF CB OZ*

🦋 Reading Assignment:

- ☞ Read about **Vernam Cipher**: About History, Application, Algorithm and related issues.
- ☞ Note: You are supposed to include this encryption algorithm in your project.

SUBSTITUTION CIPHER - POLYALPHABETIC

Hill Cipher

- A polyalphabetic cipher invented by Lester S. Hill & plaintext is divided into equal size blocks
- The blocks are encrypted one at a time and makes it a block cipher
- Each character in the block contains to the encryption of other characters in the block
- In Hill cipher, the key is a square matrix of size **m x m** in which m is the size of the block
- Let the key be K, each element of the matrix is K_{ij} as shown below

$$\text{➤ } \mathbf{k} = \begin{vmatrix} k_{11} & k_{12} & \dots & k_{1m} \\ k_{21} & k_{22} & \dots & k_{2m} \\ \vdots & \vdots & & \vdots \\ k_{m1} & k_{m2} & \dots & k_{mm} \end{vmatrix}$$

$$C_1 = P_1K_{11} + P_2K_{21} + \dots + P_mK_{m1}$$

$$C_2 = P_1K_{12} + P_2K_{22} + \dots + P_mK_{m2}$$

....

$$C_m = P_1K_{1m} + P_2K_{2m} + \dots + P_mK_{mm}$$

SUBSTITUTION CIPHER - POLYALPHABETIC

Hill Cipher

- ☞ The substitution is determined by m linear equations in which each character is assigned a numerical value ($a = 0, b = 1 \dots z = 25$)
- ☞ For $m = 3$, the system can be described as follows:

$$\begin{pmatrix} c1 \\ c2 \\ c3 \end{pmatrix} = \begin{pmatrix} k11 & k12 & k13 \\ k21 & k22 & k23 \\ k31 & k32 & k33 \end{pmatrix} \begin{pmatrix} p1 \\ p2 \\ p3 \end{pmatrix}$$

$$C1 = (k11P1 + k12P2 + k13P3) \bmod 26$$

$$C2 = (k21P1 + k22P2 + k23P3) \bmod 26$$

$$C3 = (k31P1 + k32P2 + k33P3) \bmod 26$$

- ☞ Example: Given a plaintext = PAY MORE MONEY and the encryption key as follow.

$$K = \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 12 \\ 2 & 2 & 19 \end{pmatrix} \text{ Find the encryption text.}$$

Solution: ***LNS HDL EWMTRW***

SUBSTITUTION CIPHER - POLYALPHABETIC

🐦 Hill Cipher

- ☞ Decryption requires using inverse of the matrix K .
- ☞ The Inverse K^{-1} of a matrix K is defined by the equation $KK^{-1} = K^{-1}K = I$ where I is the identity matrix
- ☞ The Hill system can be expressed as:

$$C = E(K, P) = KP \bmod 26$$

$$P = D(C, K) = K^{-1}C \bmod 26 = K^{-1}KP$$

- ☞ Hill cipher completely hides single-letter frequencies
- ☞ Hill cipher is strong against a cipher text-only attack, it is easily broken with a known plaintext attack

REVIEW EXERCISES

- ✎ Encrypt the message “meet me at the usual place at ten rather than eight o’clock” using the Hill cipher with the key $\begin{pmatrix} 9 & 4 \\ 4 & 7 \end{pmatrix}$. Show your calculations and the result. Show the calculations for the corresponding decryption of the ciphertext to recover the original plaintext.
- ✎ Using the Vigenère cipher, encrypt the word “explanation” using the key leg.
- ✎ Write a program that can perform a letter frequency attack on an additive cipher without human intervention. Your software should produce possible plaintexts in rough order of likelihood. It would be good if your user interface allowed the user to specify “give me the top 10 possible plaintexts.”
- ✎ NB: All these exercises and more exercises can be found on the text book.